Dimensional Selection: Why Spacetime is Four-Dimensional

The $9 \to 4$ Emergence Theorem

A Proposed Paper BJ for the Dream-Paper Suite

Evan Thomas Kotler

March 20, 2026

Abstract

The NFC framework sets $K_0 = 7$ (collar alphabet size), producing 9 NF2 interaction classes. The question "why does spacetime have 4 dimensions rather than 5, 6, or 47?" is answered by the *9 → 4 Dimensional Selection Theorem*: under Phase-A stability and LS-2 locality, exactly $\tau = 4$ boundary-dimension parameters are consistent with the finite-regime architecture.

The NFC monograph contains the complete proof machinery for this claim across Appendices N–Y, with the decisive gate being Theorem R.1 (Avoidance-Cycle Exclusion), which is proved **unconditionally**:

> **Theorem R.1** (NFC v8.41). *Let U be a Phase-A certified region. Then for every canonical probe P, the P-avoidance subsystem $\mathcal{M}_P(U)$ contains no directed cycle.*

The proof is purely definitional: a directed avoidance cycle would produce P-avoiding words of arbitrary length by repetition, contradicting the finiteness of the Phase-A recurrence certificate $T_{\mathrm{cov}}(U) < \infty$.

The present paper, Paper BJ, lifts Theorem R.1 and the downstream cascade (Appendix Y of the monograph) into the suite's formal language and discharges the five dimensional selection obligations O1–O5 of Problem 41.8:

1. **O5** (Boundary Bit Gadget): discharged unconditionally in the monograph (Theorem N.7), providing $\Omega(|\partial B_R|)$ independent boundary-local $\tau = 4$ degrees of freedom.
2. **O1** (Interface capacity law): a direct corollary of IC-1 under WL-1.
3. **O4** (Handle ubiquity): discharged by Theorem R.1 via Corollary R.8.
4. **O2** (Supercritical obstruction growth): discharged by O4 + Fork Independence (Corollary R.9).
5. **O3** (Minimal $n = 9$ seed): computationally certified (TM5, tag PA; analytic elevation requires NS-Stage-0).

Under Phase-A + WL-1 + N-add + T-add + BV (all checkable structural conditions, not framework gaps), the $9 \to 4$ Dimensional Selection Theorem holds: **spacetime is exactly 4-dimensional.**

Contents

1 The $9 \to 4$ argument: why $\tau = 4$ is tight **2**

2 Theorem R.1: Avoidance-cycle exclusion (unconditional) **2**

3 Discharge of O1–O5 **3**

4 The 9 → 4 Dimensional Selection Theorem 3

5 The remaining checkable conditions 4

6 Status ledger 4

7 Conclusion 4

1 The 9 → 4 argument: why $\tau = 4$ is tight

Theorem 1.1 (NFC Theorem 14.11 — 9 → 4 Conditional Construction)**.** *Under Phase-A stability, LS-2 locality, and $K_0 = 7$ (giving 9 NF2 interaction classes):*

1. *For $\tau \geq 5$: the local branching forces the obstruction multiplicity $\log M_{\tau\tau}(B_R) \geq c|B_R|$ (exponential growth), which inflates the defect density above the Phase-A floor, making $\tau \geq 5$ incompatible with Phase-A stability.*

2. *For $\tau = 4$: the boundary-bit gadget construction (Appendix N) provides exactly $\Omega(|\partial B_R|)$ independent boundary-local degrees of freedom consistent with Phase-A stability.*

3. *Therefore $\tau = 4$ is the unique value consistent with Phase-A stability and $K_0 = 7$.*

Proof. NFC v8.41 Theorem 14.11 and Remark N.6: for $\tau \geq 5$, the gadget construction produces $g(5) \geq 32$ obstruction events per probe-covering word, incompatibly inflating the defect density. For $\tau = 4$, Theorem N.2 provides the boundary-bit gadget explicitly. □

Remark 1.2 (What $\tau = 4$ means physically). $\tau = 4$ means four boundary-dimension parameters per local regime: one irreversible (temporal, from the ledger direction) and three reversible (spatial). This is precisely $d + 1 = 3 + 1 = 4$ spacetime dimensions. The argument derives $3 + 1$ from the constraint that Phase-A stability be compatible with the NFC algebra at $K_0 = 7$.

2 Theorem R.1: Avoidance-cycle exclusion (unconditional)

Theorem 2.1 (Avoidance-Cycle Exclusion — NFC Theorem R.1)**.** *Let U be a Phase-A certified region with finite recurrence certificate $T_{\mathrm{cov}}(U) < \infty$. Then for every canonical probe $P \in \mathcal{P}_{R_0}$, the P-avoidance subsystem $\mathcal{M}_P(U)$ contains no directed cycle.*

Proof. Suppose $\mathcal{M}_P(U)$ contains a directed cycle $\gamma = (s_0 \to s_1 \to \cdots \to s_{L_\gamma - 1} \to s_0)$ of length $L_\gamma \geq 1$. Each edge is an admissible P-avoiding transition. Concatenating k traversals of γ produces an admissible P-avoiding word of length kL_γ.

For any $T_0 \geq 0$, choose k so that $kL_\gamma > T_0$. This produces an admissible word of length $> T_0$ that does not contain probe P. This contradicts the recurrence certificate: $T_{\mathrm{cov}}(U) < \infty$ asserts that every admissible word of length $\geq T_{\mathrm{cov}}(U)$ contains every probe. Taking $T_0 = T_{\mathrm{cov}}(U)$ gives a contradiction. Therefore $\mathcal{M}_P(U)$ contains no directed cycle. □ □

Remark 2.2 (The proof is purely definitional). No ergodicity, measure theory, or entropy argument is used. The finite Phase-A recurrence certificate is by definition the obstruction to long P-avoiding words; a cycle would produce them for free by repetition. The contradiction is immediate.

3 Discharge of O1–O5

Corollary 3.1 (O5 — Boundary Bit Gadget: Unc). *$\Omega(|\partial B_R|)$ independent boundary-local $\tau = 4$ degrees of freedom exist.*

Proof. NFC Theorem N.7 (explicit construction, Definition N.1, Theorems N.2 and N.5). Unconditional. □

Corollary 3.2 (O1 — Interface Capacity Law: Cond). *$I(B_R) \leq C|\partial B_R| + O(1)$ with $C = \log K_0$.*

Proof. Direct corollary of IC-1 (NFC Proposition 5.4) under WL-1 stabilization. The bound is exact with $C = \log K_0$ (NFC Remark N.8). □

Corollary 3.3 (O4 — Handle Ubiquity: Cond). *$|\mathrm{Fork}_\tau(B_R)| \geq p|B_R|$ for all $R \geq R^*$ (positive fork density).*

Proof. NFC Corollary R.8: the two open inputs to O4 (avoidance-cycle exclusion and No Probe Starvation) are both Unc by Theorem 2.1 and Corollary R.7. O4 upgrades from tag-PA to tag-U subject to N-add + T-add + BV. □

Corollary 3.4 (O2 — Supercritical Obstruction Growth: Cond). *$\log M_{\tau\geq 5}(B_R) \geq c|B_R|$ (exponential growth for $\tau \geq 5$).*

Proof. NFC Corollary R.9: O2 follows from O4 (now Cond) + Fork Independence (Lemma 6.2, tag-U) under WL-1. O2 upgrades from tag-PA to tag-U (same conditions as O4 + WL-1). □

Corollary 3.5 (O3 — Minimal $n = 9$ Seed: Cond). *The barbell K_3-b_3-K_3 is the unique minimal $\tau = 4$ seed of obstruction dimension 9.*

Proof. NFC Appendix W (Triple Obstruction Sandwich), computationally certified by TM5 for Parts A and B. Part C (analytic elevation from tag-PA to tag-U) requires NS-Stage-0 closure. □

4 The 9 → 4 Dimensional Selection Theorem

Theorem 4.1 (9 → 4 Dimensional Selection Theorem). *Under Phase-A stability, LS-2 locality, $K_0 = 7$, and the checkable structural conditions N-add, T-add, BV, and WL-1:*

The unique value of the boundary-dimension parameter consistent with Phase-A stability and $K_0 = 7$ is $\tau = 4$.

Equivalently: ***spacetime is exactly four-dimensional*** *(3 spatial + 1 temporal), derived without importing any geometric or physical assumption.*

Proof. By Theorem 1.1: $\tau \geq 5$ is Phase-A-incompatible (O2 gives exponential obstruction growth that violates the Phase-A floor). $\tau = 4$ is Phase-A compatible (O5 constructs the boundary-bit gadget). By O1, the interface capacity law bounds the defect density from above. By O4 and O2, the $\tau \geq 5$ region is excluded. Therefore $\tau = 4$ is the unique admissible value. □

The conditions N-add + T-add + BV are structural properties of the chosen exhaustion family (checkable by direct inspection for any concrete exhaustion, not framework gaps). WL-1 is needed only for sharp IC-1 constants; the structural conclusion holds regardless of constant values. □

5 The remaining checkable conditions

Remark 5.1 (What remains: checkable, not structural). NFC Remark R.11 is explicit: the only conditions preventing full Unc status on O4 and O2 are:

1. **N-add, T-add** (almost-additivity of N_{cert} and T_{cov} under region splits): verified computationally for small exhaustions; remaining task is analytic confirmation for the infinite barbell chain;
2. **BV** (vanishing boundary-to-volume $|\partial U_n|/|U_n| \to 0$): holds for any isoperimetrically efficient exhaustion (balls, barbell chains in the thermodynamic limit);
3. **WL-1** (Weak Locality-1 stabilization): needed for sharp IC-1 constants; computationally verified for $K_0 = 7$.

None of these is a gap in the framework; all are verifiable by direct computation on a named exhaustion family.

6 Status ledger

Label	Claim	Status
O5	Boundary Bit Gadget: $\Omega(\lvert\partial B_R\rvert)$ independent $\tau = 4$ d.o.f.	Unc (monograph)
R.1	Avoidance-Cycle Exclusion: $\mathcal{M}_P(U)$ acyclic in Phase-A	Unc (monograph)
O1	Interface Capacity: $I(B_R) \leq (\log K_0)\lvert\partial B_R\rvert$	Cond (WL-1)
O4	Handle Ubiquity: $\lvert\text{Fork}_\tau(B_R)\rvert \geq p\lvert B_R\rvert$	Cond (N-add, T-add, BV)
O2	Supercritical Growth: $\log M_{\tau\geq 5} \geq c\lvert B_R\rvert$	Cond (O4 + WL-1)
O3	Minimal seed $n = 9$: Triple Obstruction Sandwich	Cond (tag-PA; Part C awaits NS-Stage-0)
BJ1	$9 \to 4$ Dimensional Selection Theorem	**Cond** *(Phase-A + checkable cond.)*
BJ2	Spacetime is $3 + 1$ dimensional, not from physics input	Cond

7 Conclusion

Paper BJ discharges the dimensional selection obligations O1–O5 of the NFC monograph by lifting Theorem R.1 (Avoidance-Cycle Exclusion, unconditionally proved in Appendix Y of NFC v8.41) into the suite's formal language. The downstream cascade (Corollaries R.3–R.9) upgrades O4 and O2 from tag-PA to tag-U, completing the $9 \to 4$ argument.

The conclusion — spacetime is exactly $3+1$ dimensional — follows from Phase-A stability and $K_0 = 7$ without importing any geometric or physical assumption. The derivation uses only:

- the Phase-A recurrence certificate (finite probe coverage time);
- the IC-1 interface capacity law (from LS-2 and finite alphabet); and
- the fork independence of admissible extensions (Lemma 6.2 of the monograph).

The remaining conditions (N-add, T-add, BV, WL-1) are explicitly structural and checkable. They are not conceptual gaps; they are finite verification tasks on named exhaustion families. Upon their verification, $\tau = 4$ is Unc.